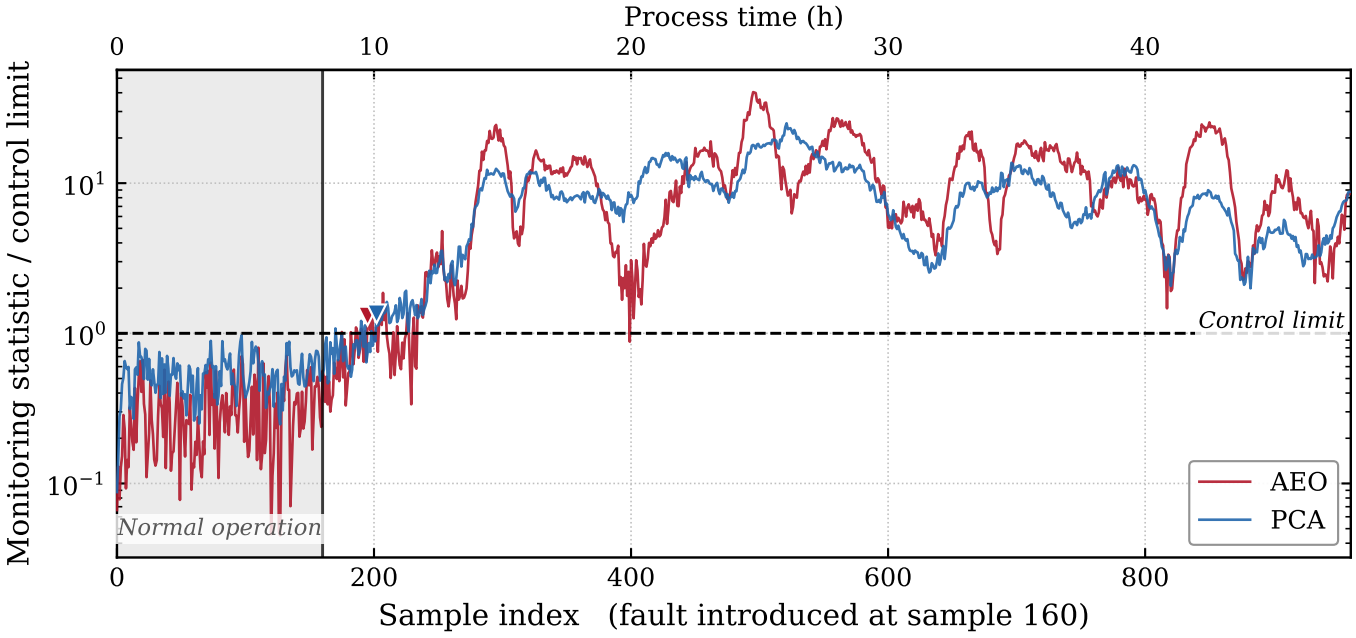
Fault Detection Performance for IDV(13)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	48.7 samples (146 min)	MDR 0.081	detected in 7/7 runs
PCA	delay	45.3 samples (136 min)	MDR 0.060	detected in 7/7 runs

PCA detects 3.4 samples (10 min) earlier; PCA missed-detection rate lower by 26%